

2):

1. DEF: 1-dim Gaussian variable
2. Apparently weaker definition of 1-dim Gaussian variable
3. Moments of BM^1
4. DEF: 1-dim Gaussian process
5. THM: X stoc proc. Then tfae:
 - $X \text{ } BM^1$
 - X 1-dim Gaussian process with cov matr ..., mean vector 0 and continuous paths
6. Invariance Closure of BM^1 :
 - Reflection
 - Renewal + independence of rwn process at a from original proc stopped at a
 - Time inversion
 - Scaling
7. DEF: Independence of processes
8. THM: $\vec{X}$ stoc proc. tfae:
 - $\vec{X}$ satisfies BM0,...,3
 - $\forall n < \omega \forall 0 = t_0 < \dots < t_n \forall \vec{\xi}_0, \dots, \vec{\xi}_n \in \mathbb{R}^d \mathbb{E}[\exp(i \sum_{j=1, \dots, n} \vec{\xi}_j \cdot (\vec{X}_{t_j} - \vec{X}_{t_{j-1}})) + i \vec{\xi}_0 \cdot \vec{X}_{t_0}] = \exp(-\frac{1}{2} \sum_{j=1, \dots, n} \vec{\xi}_j^T \cdot (t_j - t_{j-1}) Id \cdot \vec{\xi}_j)$
9. THM: $\vec{X}$ stoc proc. tfae:
 - $\vec{X} \text{ } BM^d$
 - $\vec{X}^1, \dots, \vec{X}^d$ independent BM^1